计算机网络课程设计

设计开发DNS Relay

感谢北邮计算机学院高占春副教授提供原版课件

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课程介绍

- 本学期通信与网路课程设计为1.0学分，教学内容为综合课实验，教学方式采用开放式实验教学模式，第2周上课。
- 以目前广泛使用的TCP/IP协议栈为基础，使用UDP或TCP通信平台，利用Socket编程界面，实现DNS中继器软件或TFTP文件传送协议（Trivial File Transfer Protocol）服务器端软件。通过本课程的学习，使学生熟练掌握Socket编程界面的使用、能够对通信协议进行描述和设计、掌握计算机网络协议设计、编程和测试的基本方法和相关技术，提高计算机网络与通信系统的设计与开发和实际环境中设计开发系统的能力。

参考书

- 是学习Python语言的入门参考书，浅显易懂，值得推荐。

- 在IEEE Spectrum 2025编程语言排行榜上，Python再次斩获最受欢迎的编程语言。

- 使用Python语言的人数众多，据说在AI学习领域的开发很多使用的都是Python，未来的明日之星看好Python。

课程安排

- 基本要求
- DNS是什么
- 如何使用Wireshark软件
- 如何进行Client-Server Socket通信软件开发

DNS Relay

- 从DNS Client接收到DNS查询请求并转发DNS查询请求到一个给定的DNS Server
- 从DNS Server接收到DNS查询响应并转发给DNS解析器（DNS Resolver）

Cases

- 查找本地数据库（例如dnsrelay.txt）
  - Case 1：对于IP地址0.0.0.0, 发送返回“no such name”（reply code = 0011）
  - Case 2：对于包括在本地数据库里的domain name，发送返回相应的IP地址
  - Case 3：对于没有包括在本地数据库里的domain name，转发查询请求给本地的DNS服务器（DNS Server）

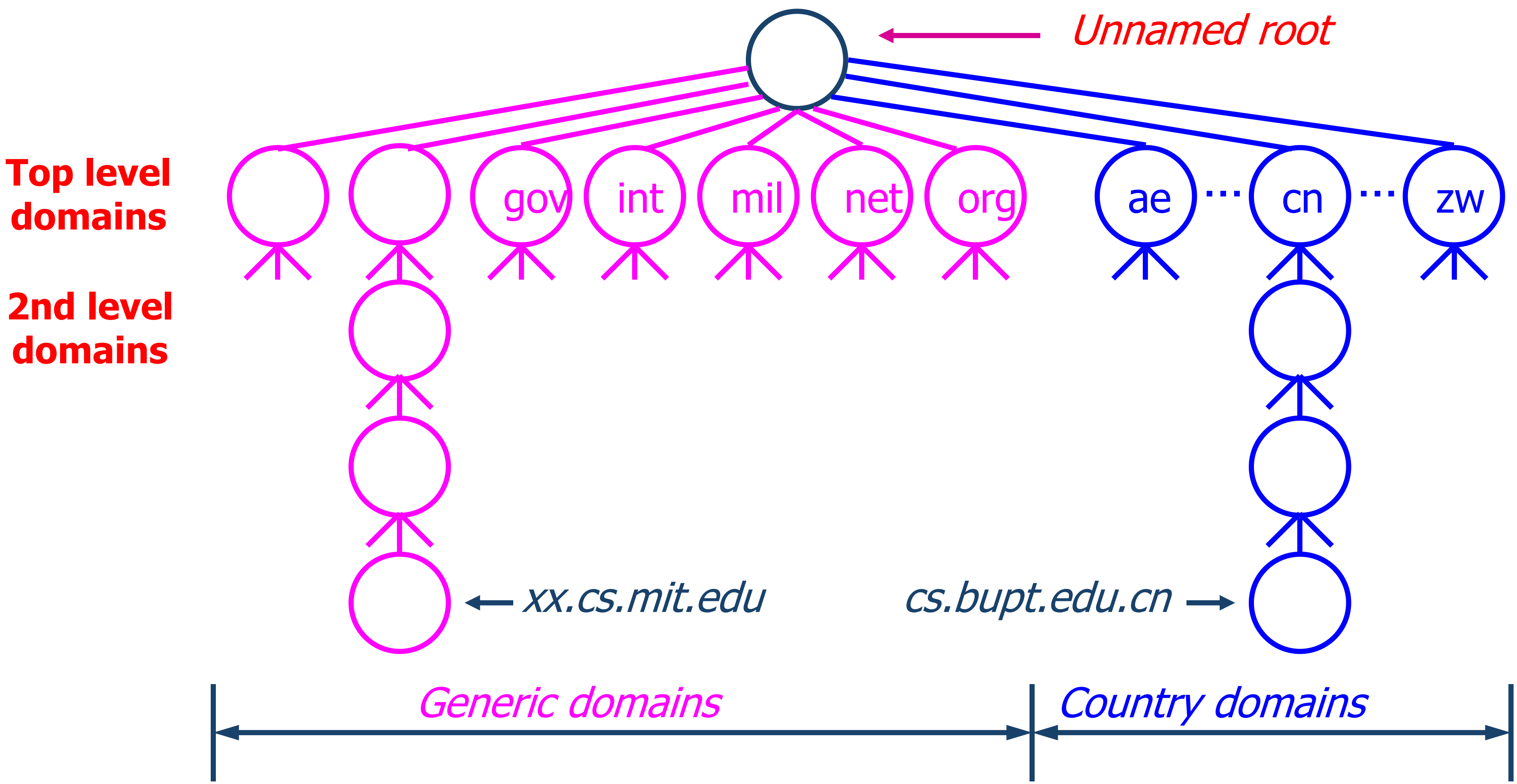
软件环境

- 操作系统（Operating Systems）
  - Windows, Linux
- 编程语言
  - Python, C/C++，Java, ...
- 工作小组
  - 每个小组三个人
  - 一个源程序和一份实验报告

开始了解DNS

- Domain Name System
- 分布式数据库文件提供域名（Domain Name）和IP地址之间的联结映射
  - Implemented in hierarchy of many name servers
- 应用层协议
  - 由hosts和Name servers通信交互来解析域名（Domain Name）
- 多在传输层（transport layer）使用UDP协议

Domain Namespace-分层结构



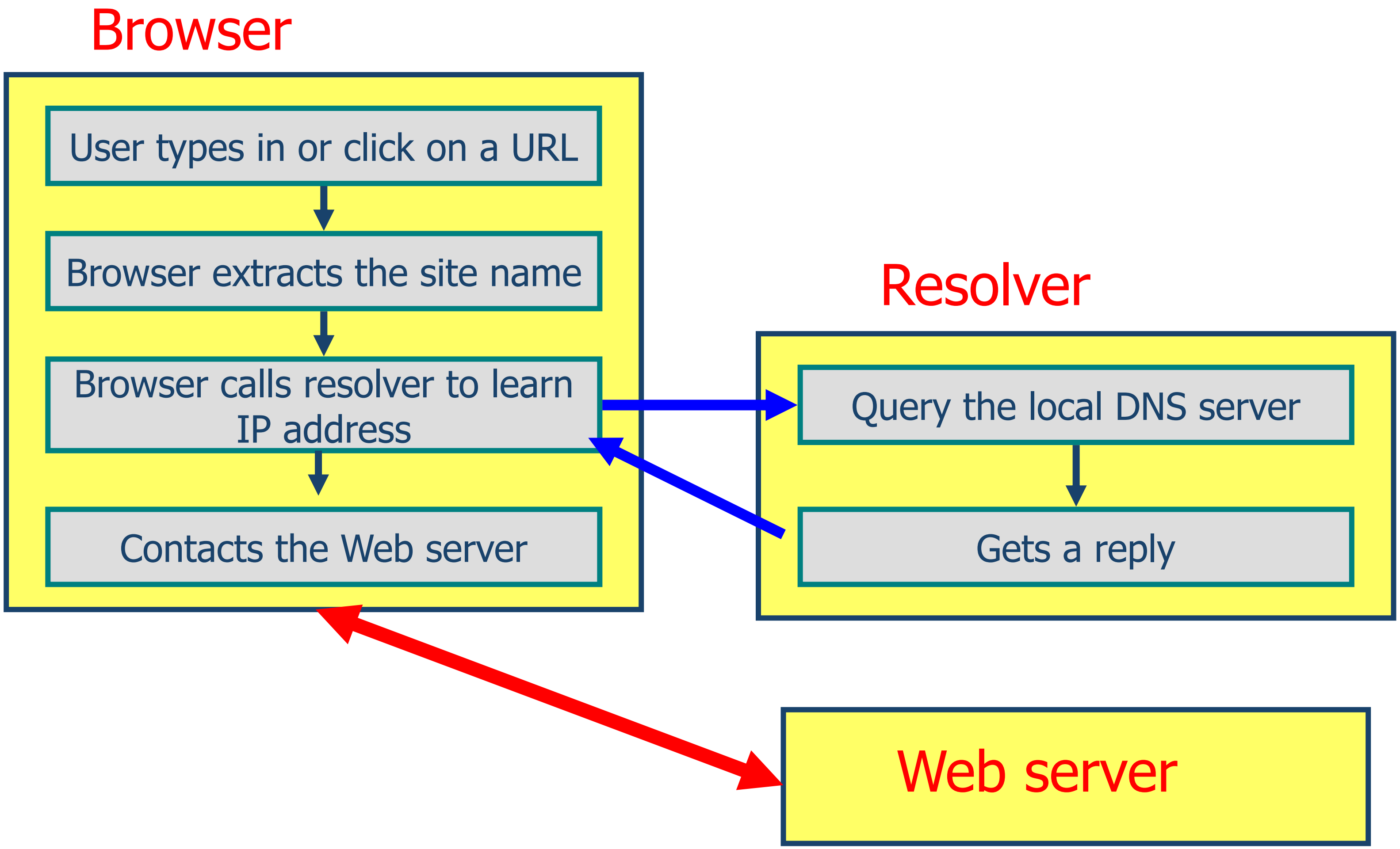
Name Servers

- Name Servers是组成域名数据库（Domain Database）的信息容器
- 回答关于域名信息的查询请求

DNS Client：解析器（Resolvers）

- 是用户程序和域名服务器之间的接口
  - resolver接收来自用户程序的请求（例如，http、mail programs、telnet、ftp、ping）
  - Asks questions to the DNS system on behalf of the application
  - 返回想要的信息

举例：DNS和HTTP application交互

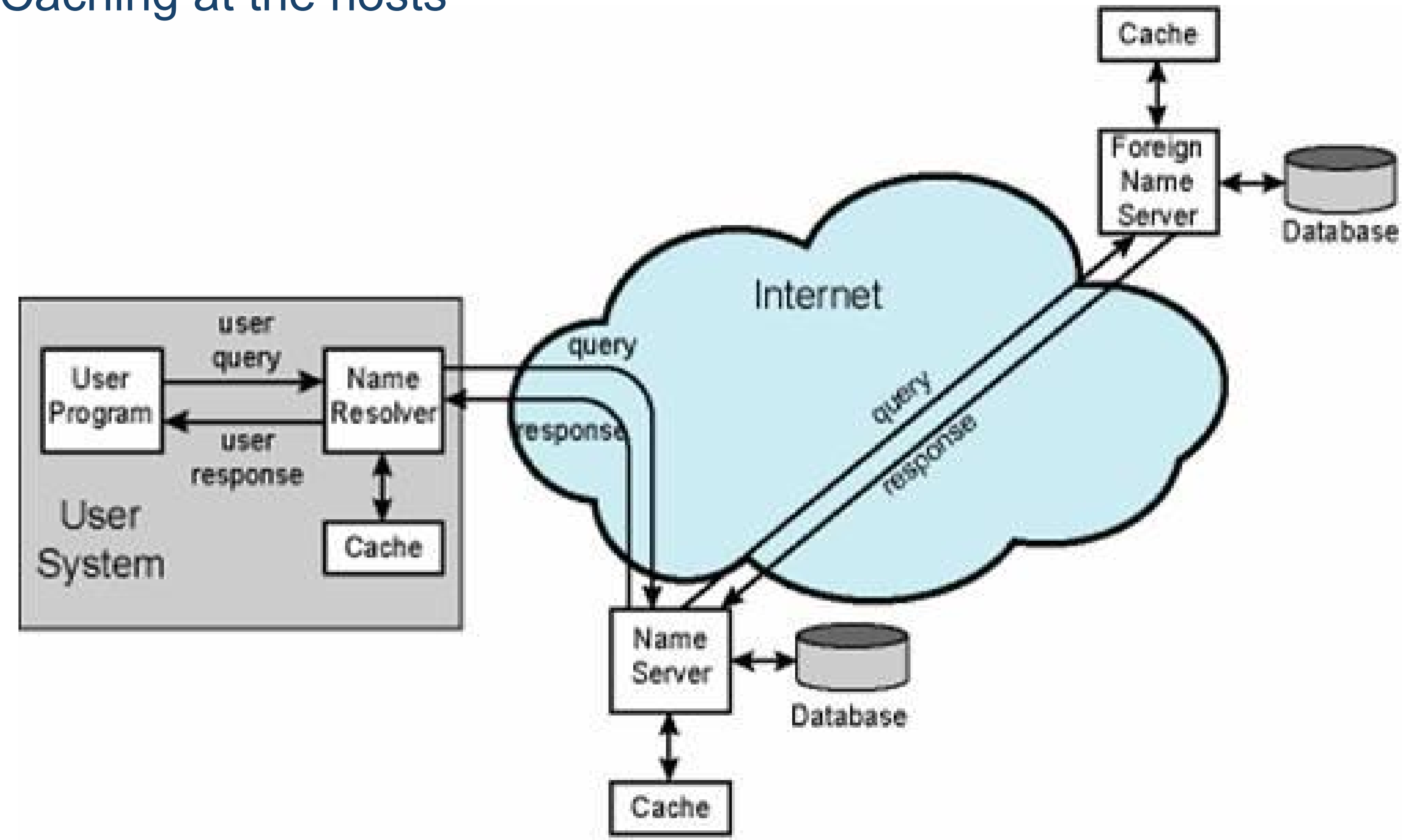


Name information: Resource Records

- **DNS中的每一个域（Domain）由一个或多个Resource Records (RRs),** 包含了域名中的字段信息
- **每个Resource Records**包含以下信息：
  - Owner: the domain name where the RR is found
  - Type: specifies the type of the resource in this RR
    - A – Host Address
    - MX – Mail Exchanger
    - ...
  - Class: specifies the protocol family to use
    - IN – the Internet system
  - TTL: specifies the Time To Live (in unit of second) of the cached RRs
  - RDATA: the resource data

Caching机制

- Caching at the name servers
- Caching at the hosts



协议细节：消息格式（根据RFC 1035）

<table>
<tr><th></th><th>0 … 15</th><th colspan="8">16 … Flags … 31</th></tr>
<tr><td rowspan="3">Fixed length</td><td>ID</td><td>QR</td><td>OPCODE</td><td>AA</td><td>TC</td><td>RD</td><td>RA</td><td>Z</td><td>Rcode</td></tr>
<tr><td>Question count</td><td colspan="8">Answer count</td></tr>
<tr><td>Authority count</td><td colspan="8">Additional count</td></tr>
<tr><td rowspan="4">Variable length</td><td colspan="9">Question Section
(variable number of questions)</td></tr>
<tr><td colspan="9">Answer Section
(variable number of Resource Recordss)</td></tr>
<tr><td colspan="9">Authority Section
(variable number of Resource Recordss)</td></tr>
<tr><td colspan="9">Additional Section
(variable number of Resource Recordss)</td></tr>
</table>

DNS消息头结构(1)

- ID: 16-bit field used to correlate queries and responses.
- QR: 1-bit field that identifies the message as a query (0) or response (1).
- OPCODE: 4-bit field that describes the type of query:
  - 0: Standard query (name to address). 1: Inverse query (address to name). 2: Server status request.
- AA: Authoritative Answer. 1-bit field. When set to 1, identifies this response is made by an authoritative name server.
- TC: Truncation. 1-bit field. When set to 1, indicates the message has been truncated due to length greater than that permitted.
- RD: Recursion Desired. 1-bit field. Set to 1 by the resolver to request recursive service by the name server.
- RA: Recursion Available. 1-bit field. Set to 1 by name server to indicate recursive query support is available.
- Z: 3-bit field. Reserved for future use. Must be set to 0.

DNS消息头结构(2)

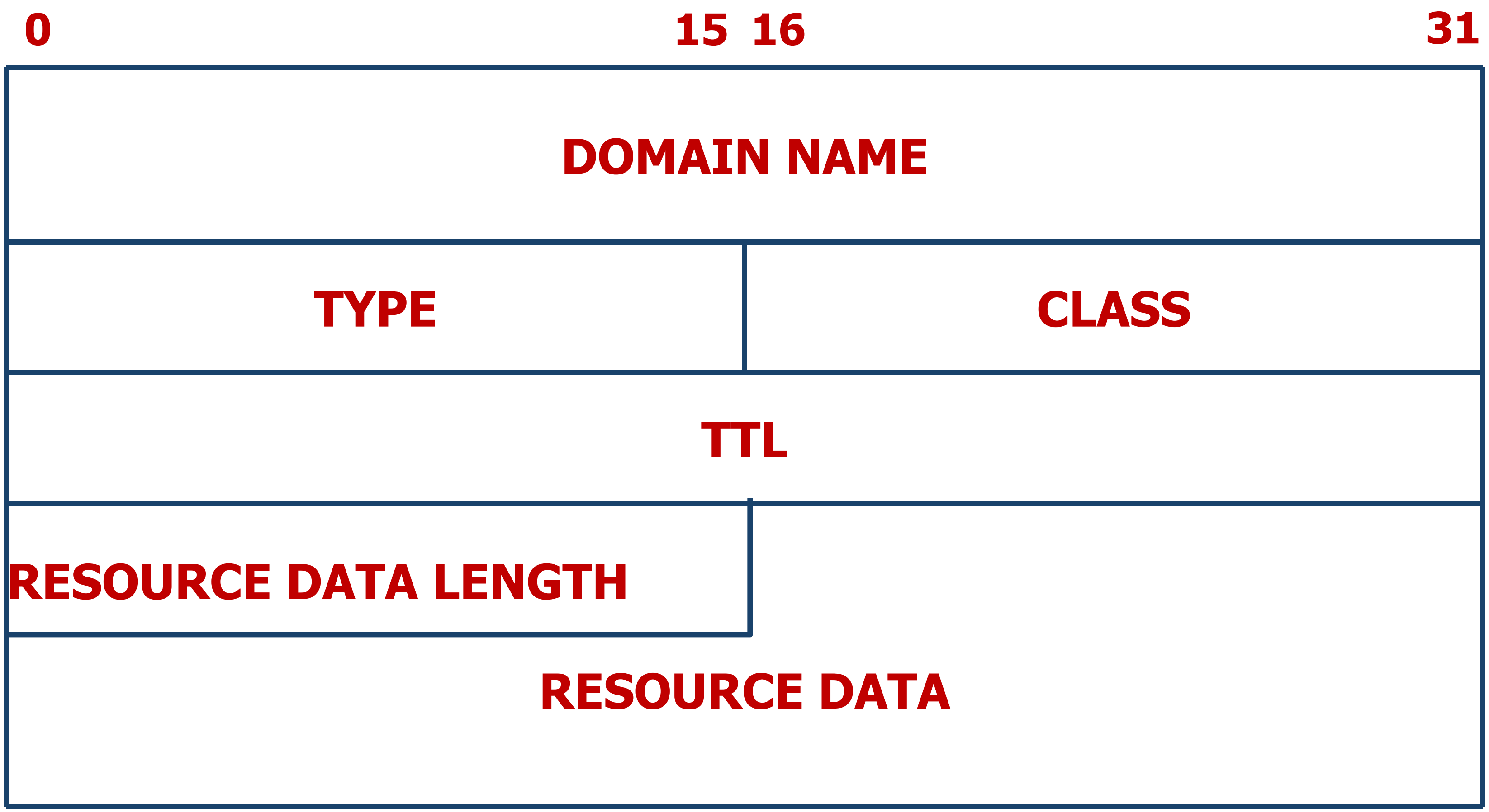
- RCODE: Response Code. 4-bit field that is set by the name server to identify the status of the query:
  - 0: No error condition. 1: Unable to interpret query due to format error. 2: Unable to process due to server failure. 3: **Name in query does not exist**. 4: Type of query not supported. 5:Query refused for policy reasons.
- QDCOUNT(Question count): 16-bit field that defines the number of entries in the question section.
- ANCOUNT(Answer count): 16-bit field that defines the number of resource records in the answer section.
- NSCOUNT(Authority count): 16-bit field that defines the number of name server resource records in the authority section.
- ARCOUNT(Additional count): 16-bit field that defines the number of resource records in the additional records section.

Question Section Format

0	15 16	31
QEURY DOMAIN NAME (variable number of domain names)		
QUERY TYPE		**QUERY CLASS**

- QUERY TYPE: 16-bit field used to specify the type of the query
  - A(1) – Host address
  - MX(15) – Mail exchanger for the domain
  - …
- QUERY CLASS: 16-bit field used to specify the class of the query
  - IN(1) – Internet system
  - …

Resource Record Format (1)



Resource Record Format (2)– type field

- **SOA**
  - Start Of Authority--identifies the domain or zone and sets a number of parameters
- **NS**
  - Maps a domain name to the name of a computer that is authoritative for the domain
- **A**
  - **Maps the name of a system to its address. If a system (e.g., a router) has several addresses, then there will be a separate record for each.**
- **AAAA**
  - **Maps the name of a system to its IPv6 address. If a system (e.g., a router) has several addresses, then there will be a separate record for each.**
- **CNAME**
  - Maps an alias name to the true, canonical name
- **MX**
  - Mail Exchanger. Identifies the systems that relay mail into the organization

Resource Record Format (3)– type field

- **TXT**
  - Provides a way to add text comments to the database. For example, a txt record could map abc.com to the company's name, address, and telephone number
- **WKS**
  - Well Known Services. Can list the application services available at the host. Used sparingly, if at all
- **HINFO**
  - Host Information, such as computer type and model. Rarely used
- **PTR**
  - Maps an IP address to a system name. Used in address-to-name files.

Example of Resource Records (1)

- Type=A
  - Name= Domain name , Value= IP Address

ns.bupt.edu.cn 86400 IN A 202.112.10.37

- **Type=NS**
  - Name= Domain, eg. bupt.edu.cn
  - value= Domain name of Authoritative Name Server

bupt.edu.cn 86400 IN NS ns.bupt.edu.cn

Example of Resource Records (2)

- Type=CNAME
  - Name= canonical name
  - Value= the name

  dns.bupt.edu.cn 86400 IN CNAME ns.bupt.edu.cn

- Type=MX
  - Name= Domain name
  - Value= canonical name of mail server

  bupt.edu.cn 86400 IN MX mail.bupt.edu.cn

DNS tools: nslookup

- Query type=A

```
命令提示符
C:\Users\DELL>nslookup   www.bupt.edu.cn
服务器:  UnKnown
Address:  10.3.9.44

非权威应答:
名称:    vn46.bupt.edu.cn
Addresses:  2001:da8:215:4038::161
          10.3.9.161
Aliases:  www.bupt.edu.cn


C:\Users\DELL>
```

- Query type=MX

```
命令提示符
C:\Users\DELL>nslookup -qt=MX mail.bupt.edu.cn
服务器:  UnKnown
Address:  10.3.9.44

非权威应答:
mail.bupt.edu.cn        canonical name = ssl.exmail.qq.com

exmail.qq.com
        primary name server = ns-tel1.qq.com
        responsible mail addr = webmaster.qq.com
        serial  = 1277374542
        refresh = 300 (5 mins)
        retry   = 600 (10 mins)
        expire  = 86400 (1 day)
        default TTL = 300 (5 mins)
```

使用Wireshark软件来理解消息通信

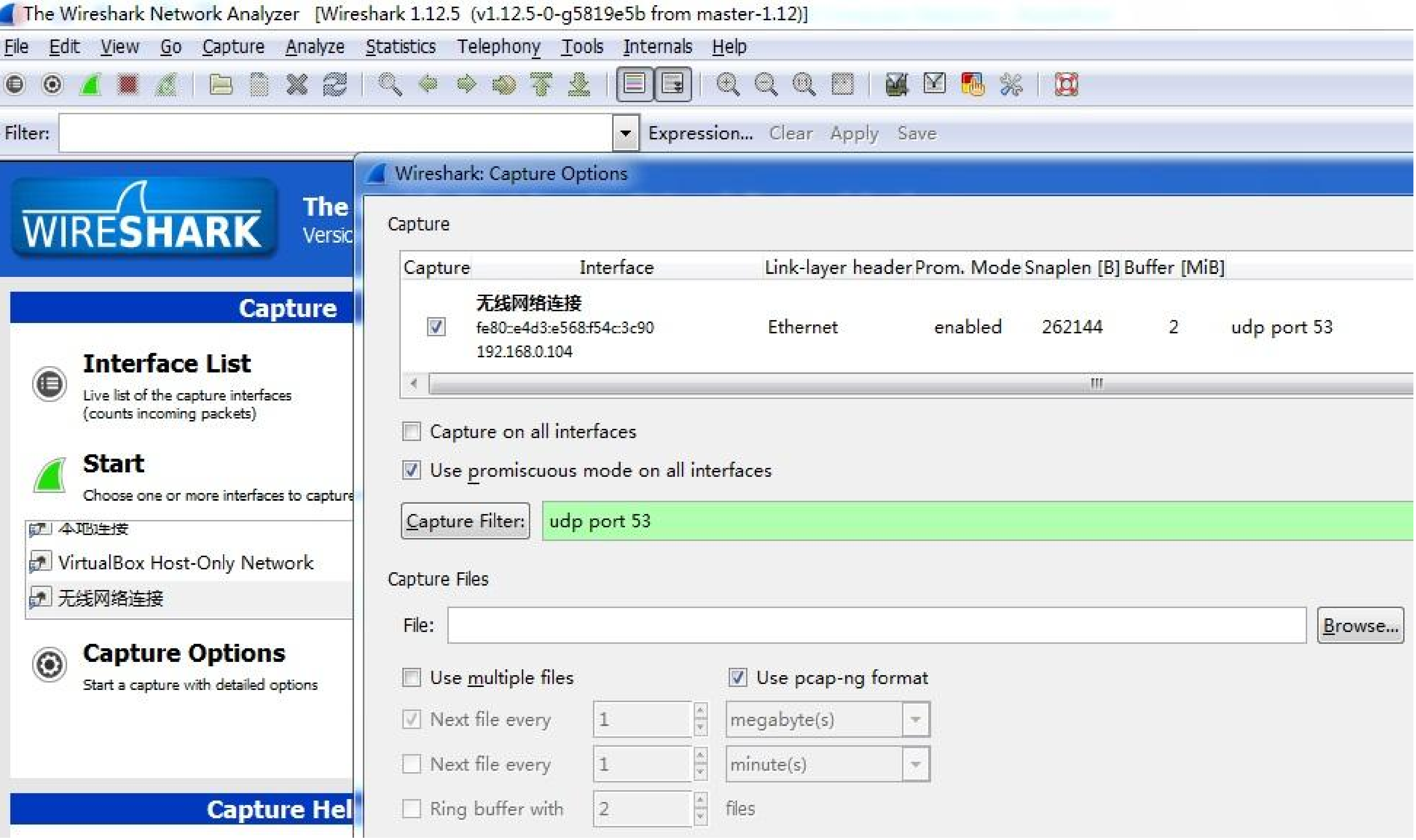
- **Wireshark: 网络协议分析器 (packet sniffer)**
- 能够在网络上抓消息包，显示消息包字段和其含义
- 用于网络故障排除、分析、软件和通信协议开发以及教育

下载Wireshark软件

- 官网网址
  - http://www.wireshark.org/
- 下载网址
  - http://www.wireshark.org/download.html

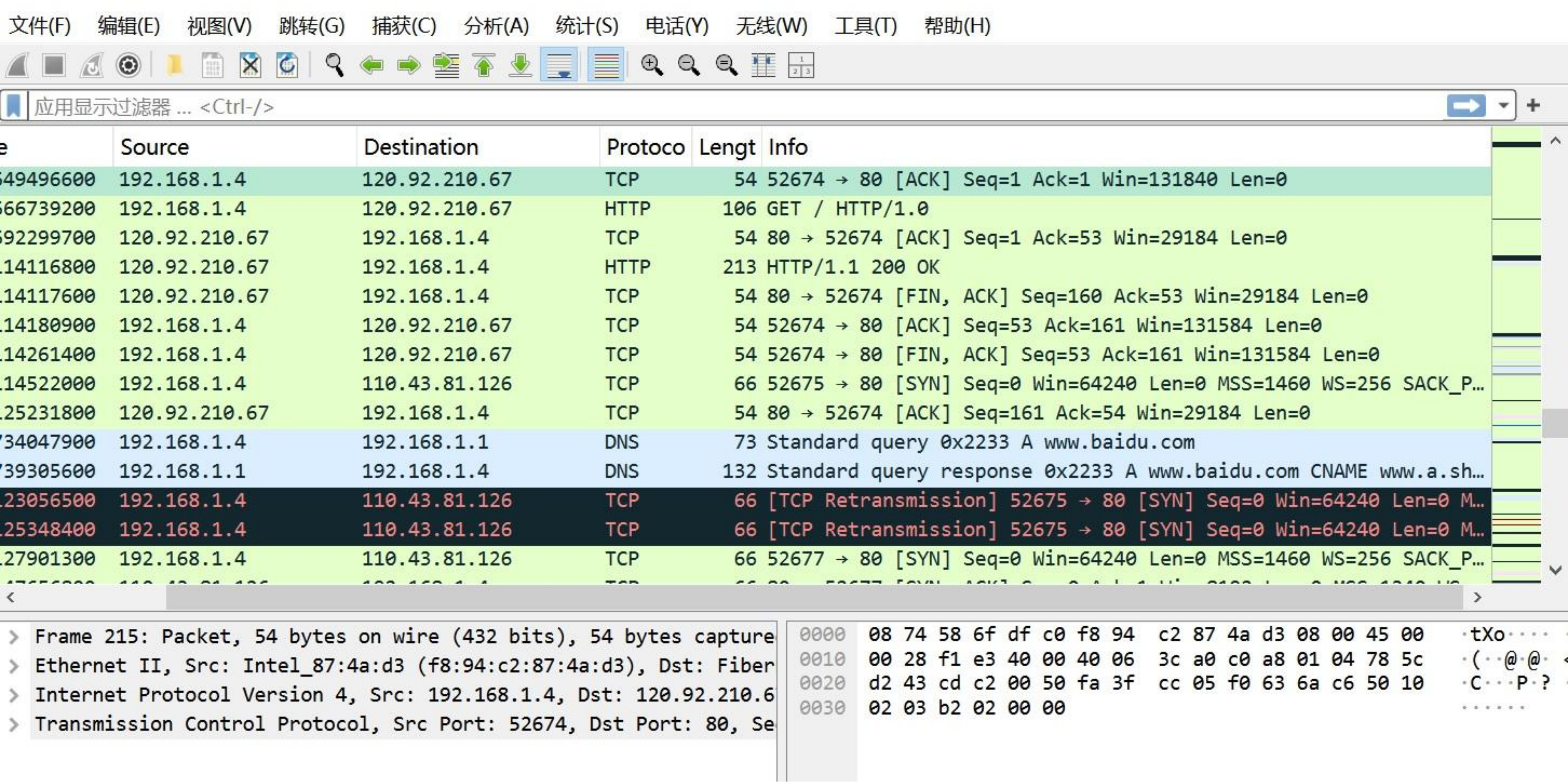
在Windows环境中Wireshark软件

以管理员身份运行，*Select Interface, set Capture Filter, then press Start*



在Windows环境中Wireshark软件

以管理员身份运行，*Select Interface, set Capture Filter, then press Start*



如何启动通信

- **DNS**
  - 启动Wireshark
  - 可以在cmd命令行环境中运行 *nslookup* 命令

DNS Request

12		W2K3-SERVER	[202.106.0.20]	DNS: C ID=34422 OP=QUERY NAME=ftp.bupt.edu.cı	75	0:00:17.779	4.061.449	2010-08-06 (
13		[202.106.0.20]	W2K3-SERVER	DNS: R ID=34422 OP=QUERY STAT=OK NAME=ftp.buı	91	0:00:17.791	0.011.661	2010-08-06 (
14		W2K3-SERVER	ftp.bupt.edu.cn	ICMP: Echo	1514	0:00:17.812	0.021.578	2010-08-06 (

```
IP:
UDP: ----- UDP Header -----
  UDP:
  UDP: Source port      = 3017
  UDP: Destination port = 53 (Domain)
  UDP: Length           = 41
  UDP: Checksum         = 3B76 (correct)
  UDP: [33 byte(s) of data]
  UDP:
DNS: ----- Internet Domain Name Service header -----
  DNS:
  DNS: ID = 34422
  DNS: Flags = 01
  DNS: 0... .... = Command
  DNS: .000 0...  = Query
  DNS: .... ..0. = Not truncated
  DNS: .... ...1 = Recursion desired
  DNS: Flags = 0X
  DNS: ...0 .... = Non Verified data NOT acceptable
  DNS: Question count = 1, Answer count = 0
  DNS: Authority count = 0, Additional record count = 0
  DNS:
  DNS: ZONE Section
  DNS:     Name = ftp.bupt.edu.cn
  DNS:     Type = Host address (A,1)
  DNS:     Class = Internet (IN,1)
  DNS:
```

```
00000000: 00 23 cd 82 e0 f6 00 0f 1f 52 ef f6 08 00 45 00  .#蚺圄...R秭..E.
00000010: 00 3d 01 82 00 00 80 11 ad ff c0 a8 00 08 ca 6a  .=.?.€.?括..蕒
00000020: 00 14 0b c9 00 35 00 29 3b 76 86 76 01 00 00 01  ...?5.);v 啾....
00000030: 00 00 00 00 00 00 03 66 74 70 04 62 75 70 74 03  .......ftp.bupt.
00000040: 65 64 75 02 63 6e 00 00 01 00 01                 edu.cn.....
```

DNS Response

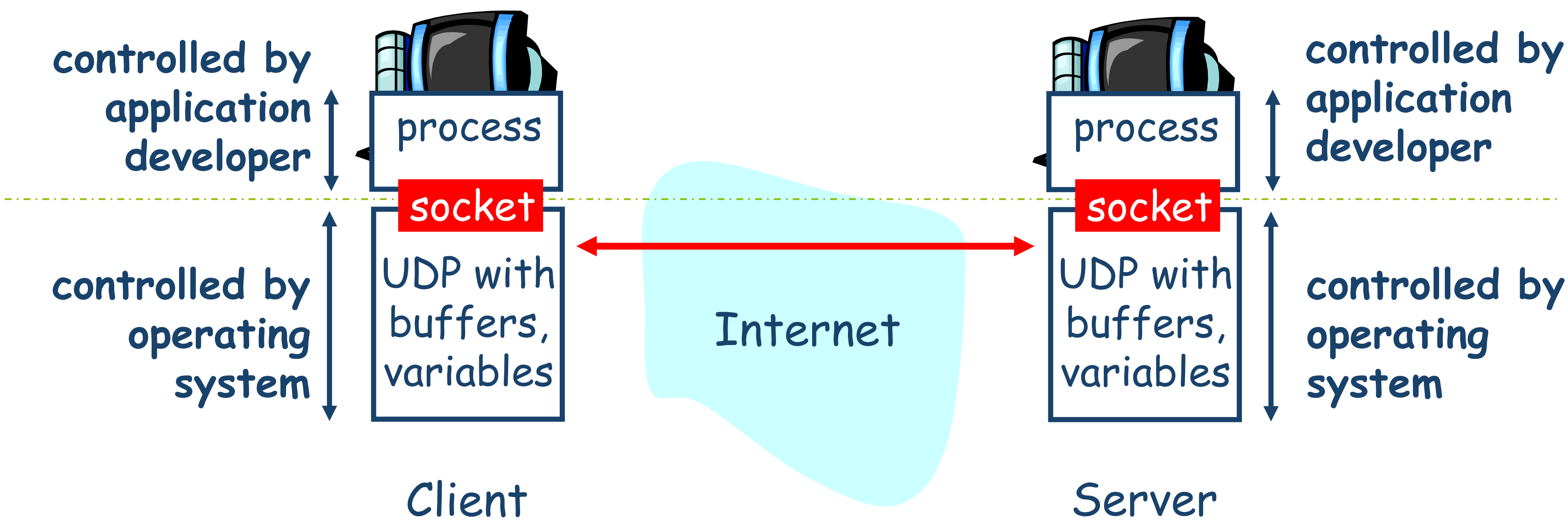
```
13  [202.106.0.20]  W2K3-SERVER     DNS: R ID=34422 OP=QUERY STAT=OK NAME=ftp.bu|  91    0:00:17.791  0.011.661  2010-08-06
14  W2K3-SERVER     ftp.bupt.edu.cn ICMP: Echo                                     1514  0:00:17.812  0.021.578  2010-08-06
15  W2K3-SERVER     ftp.bupt.edu.cn IP: Continuation of frame 14; 348 Bytes of da  362   0:00:17.812  0.000.085  2010-08-06

UDP:
DNS: ----- Internet Domain Name Service header -----
  DNS:
  DNS: ID = 34422
  DNS: Flags = 81
  DNS: 1... .... = Response
  DNS: .... .0.. = Not authoritative answer
  DNS: .000 0...   = Query
  DNS: .... ..0. = Not truncated
  DNS: Flags = 8X
  DNS: ..0. .... = Data NOT verified
  DNS: 1... .... = Recursion available
  DNS: Response code = OK (0)
  DNS: ...0 .... = Unicast packet
  DNS: Question count = 1, Answer count = 1
  DNS: Authority count = 0, Additional record count = 0
  DNS:
  DNS: ZONE Section
  DNS:     Name = ftp.bupt.edu.cn
  DNS:     Type = Host address (A,1)
  DNS:     Class = Internet (IN,1)
  DNS:
  DNS: Answer section:
  DNS:     Name = ftp.bupt.edu.cn
  DNS:     Type = Host address (A,1)
  DNS:     Class = Internet (IN,1)
  DNS:     Time-to-live = 6426 (seconds)
  DNS:     Length = 4
  DNS:     Address = [211.68.71.80]
  DNS:

00000000: 00 0f 1f 52 ef f6 00 23 cd 82 e0 f6 08 00 45 00  ...R秭.#蚺图..E.
00000010: 00 4d 29 cc 00 00 3a 11 cb a5 ca 6a 00 14 c0 a8  .M)?..:.痕胥..括
00000020: 00 08 00 35 0b c9 00 39 f8 e0 86 76 81 80 00 01  ...5.?9   啾亏..
00000030: 00 01 00 00 00 00 03 66 74 70 04 62 75 70 74 03  .......ftp.bupt.
00000040: 65 64 75 02 63 6e 00 00 01 00 01 c0 0c 00 01 00  edu.cn.....?...
00000050: 01 00 00 19 1a 00 04 d3 44 47 50                 .......緦GP
```

Client-Server socket通信编程

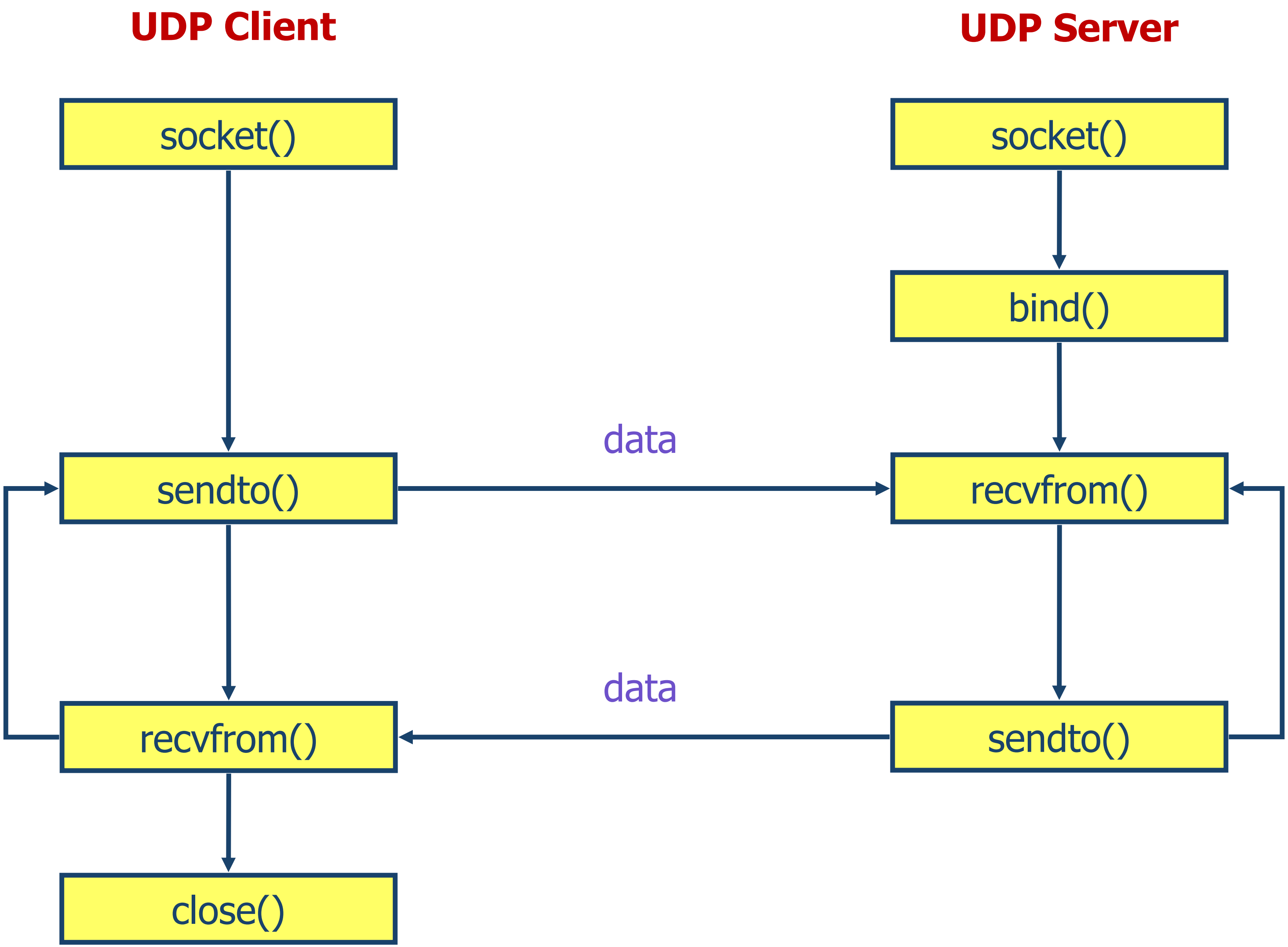
- **Socket**套接口：操作系统**I/O**系统的扩展，用于实现进程间及机器间的通信

- 一种接口（“门”），应用程序进程可通过其向另一（远程或本地）应用程序进程发送和接收消息

- **Socket套接口**：应用程序进程与端到端传输协议（**UDP**或**TCP**）之间的接口（即"门"）
- **UDP服务**：实现进程间数据包的不可靠传输



基于UDP的Socket API概述

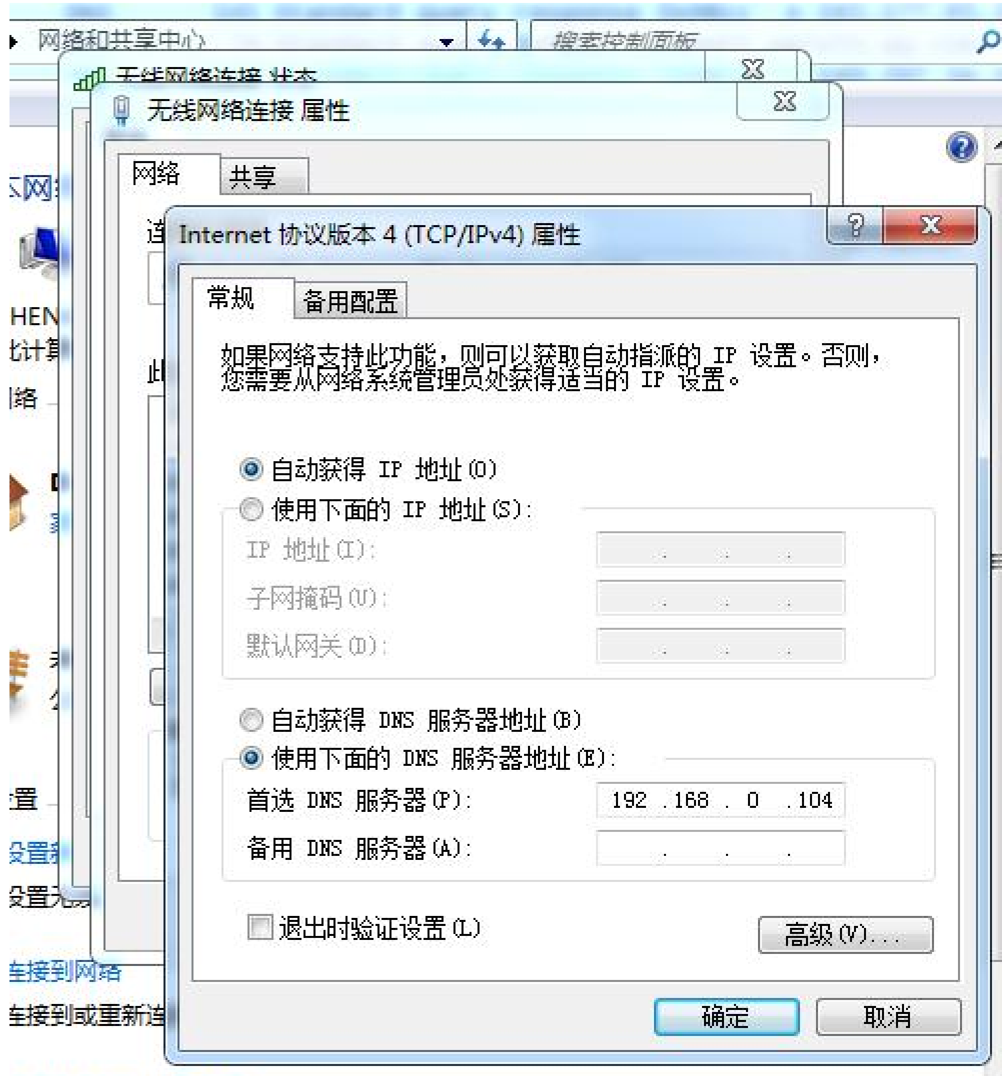


运行程序（1）

- 1. Get the IP address of your DNS server (eg. *202.106.0.20*) and your IP address (eg. *192.168.0.104*)

  c:> ipconfig /all

- 2. configure your DNS server as your IP address (eg. *192.168.0.104 or 127.0.0.1*)



运行Client-Server Socket通信程序(2)

命令提示符

```
D:\python_work>cd 10_Socket_Client-Server

D:\python_work\10_Socket_Client-Server>dir
 驱动器 D 中的卷是 新加卷
 卷的序列号是 5028-A885

 D:\python_work\10_Socket_Client-Server 的目录

2026/06/28  15:30    <DIR>          .
2026/06/28  15:30    <DIR>          ..
2026/06/28  14:29               902 client.py
2026/06/28  15:33    <DIR>          CSDN
2026/06/28  12:42    <DIR>          DeepSeek_V4
2026/06/28  14:28             1,449 server.py
               2 个文件          2,351 字节
               4 个目录 80,153,915,392 可用字节

D:\python_work\10_Socket_Client-Server>python client.py
已成功连接到服务器！输入'exit'退出连接

请输入要发送的消息：Hello,server 2026!
服务端回复：服务端已收到：Hello,server 2026!

请输入要发送的消息：Hello, July 2026!
服务端回复：服务端已收到：Hello, July 2026!

请输入要发送的消息：exit
已断开与服务端的连接

D:\python_work\10_Socket_Client-Server>_
```

搜索 明天有暴雨 英 15:48 2026/6/28

运行Client-Server Socket通信程序(2)

命令提示符 - server.py

```
2022/08/05  18:37         1,520,422 python 中如何Debug_测试小白球的博客-CSDN博客_python代码debug.png
2025/12/08  10:57            11,505 python学习心得.docx
               4 个文件      3,407,417 字节
              12 个目录 80,153,915,392 可用字节

D:\python_work>cd 10_Socket_Client-Server

D:\python_work\10_Socket_Client-Server>dir
 驱动器 D 中的卷是 新加卷
 卷的序列号是 5028-A885

 D:\python_work\10_Socket_Client-Server 的目录

2026/06/28  15:30    <DIR>          .
2026/06/28  15:30    <DIR>          ..
2026/06/28  14:29               902 client.py
2026/06/28  15:33    <DIR>          CSDN
2026/06/28  12:42    <DIR>          DeepSeek_V4
2026/06/28  14:28             1,449 server.py
               2 个文件          2,351 字节
               4 个目录 80,153,915,392 可用字节

D:\python_work\10_Socket_Client-Server>server.py
服务端已启动，正在监听端口 80 ...

客户端已连接： ('127.0.0.1', 51591)
收到客户端消息：Hello,server 2026!
收到客户端消息：Hello, July 2026!
客户端已断开连接
```

搜索　未来几天多雨　英　15:46 2026/6/28

运行DNS Relay client-server程序(2)

命令提示符

```
C:\Users\Laptop>d:

D:\>cd python_work

D:\python_work>cd DNS_relay

D:\python_work\DNS_relay>dir
 驱动器 D 中的卷是 新加卷
 卷的序列号是 5028-A885

 D:\python_work\DNS_relay 的目录

2026/06/30  14:58    <DIR>          .
2026/06/30  14:58    <DIR>          ..
2026/06/30  14:57    <DIR>          Client-Server
2026/06/30  15:36             1,206 client.py
2026/06/28  15:33    <DIR>          CSDN
2026/06/28  12:42    <DIR>          DeepSeek_V4
2026/06/30  15:16             1,705 dnsrelay.py
2021/07/04  14:52                59 dnsrelay.txt
2026/06/30  14:55             1,186 relay.py
               4 个文件          4,156 字节
               5 个目录 80,175,501,312 可用字节

D:\python_work\DNS_relay>python client.py
客户端：向DNS中继发送查询 www.baidu.com 请求
客户端收到中继返回的DNS原始报文(十六进制):
22338180000100030000000000377777705626169647503636f6d0000010001c00c0005000100000150000f0377777701610673686966656ec016c02b000100010000003c00046ef24515c02b00010
0010000003c00046ef24639

D:\python_work\DNS_relay>python client.py
客户端：向DNS中继发送查询 www.baidu.com 请求
客户端收到中继返回的DNS原始报文(十六进制):
22338180000100030000000000377777705626169647503636f6d0000010001c00c0005000100000131000f0377777701610673686966656ec016c02b000100010000001d00046ef24639c02b00010
0010000001d00046ef24515

D:\python_work\DNS_relay>dig @127.0.0.1 -p 5353 www.baidu.com
'dig' 不是内部或外部命令，也不是可运行的程序
或批处理文件。

D:\python_work\DNS_relay>_
```

英

搜索　雷电预警　英　16:15 2026/6/30

运行DNS Relay client-server程序(2)

命令提示符 - python dnsrelay.py

```
              4 个文件      3,407,417 字节
             13 个目录 80,175,501,312 可用字节

D:\python_work>cd DNS_relay

D:\python_work\DNS_relay>dir
 驱动器 D 中的卷是 新加卷
 卷的序列号是 5028-A885

 D:\python_work\DNS_relay 的目录

2026/06/30  14:58    <DIR>          .
2026/06/30  14:58    <DIR>          ..
2026/06/30  14:57    <DIR>          Client-Server
2026/06/30  15:36             1,206 client.py
2026/06/28  15:33    <DIR>          CSDN
2026/06/28  12:42    <DIR>          DeepSeek_V4
2026/06/30  15:16             1,705 dnsrelay.py
2021/07/04  14:52                59 dnsrelay.txt
2026/06/30  14:55             1,186 relay.py
               4 个文件          4,156 字节
               5 个目录 80,175,501,312 可用字节

D:\python_work\DNS_relay>python dnsrelay.py
DNS Relay running on 127.0.0.1:53, upstream DNS: 192.168.1.1:53
Receive DNS query from ('127.0.0.1', 51446)
Forward DNS response to ('127.0.0.1', 51446)
Receive DNS query from ('127.0.0.1', 57605)
Forward DNS response to ('127.0.0.1', 57605)
_
```

搜索 26°C 多云 16:14 2026/6/30

关于课程设计报告

- **Report**
  - 每个班的学委收齐课程设计报告并发送email到邮箱 ly@bupt.edu.cn